Complaint Management System

Software Requirements Specification

Project:	Complaint Management System (CMS)
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This document is a Software Requirements Specification (SRS) for a Complaint Management System developed as part of the BTech 3rd Year Software Engineering course project. It follows the ISO/IEC/IEEE 29148:2018 standard structure.

The system enables citizens, students, or customers to raise complaints, track their status, and receive resolutions through a structured workflow involving complaint registration, assignment, escalation, and closure.

Table of Contents

1. Introduction 4

1.1 Purpose 4

1.2 Scope 4

1.3 Product Perspective 5

1.4 Product Functions 5

1.5 User Characteristics 5

1.6 Limitations 6

1.7 Assumptions and Dependencies 6

1.8 Definitions 6

1.9 Acronyms and Abbreviations 6

2. Requirements 7

2.1 External Interfaces 7

2.2 Functions 7

2.2.1 User Registration and Authentication 7

2.2.2 Complaint Submission 8

2.2.3 Complaint Tracking 8

2.2.4 Complaint Assignment and Escalation 9

2.2.5 Resolution and Closure 9

2.2.6 Notifications 9

2.2.7 Admin and Reporting 10

2.3 Usability Requirements 10

2.4 Performance Requirements 11

2.5 Logical Database Requirements 11

2.6 Design Constraints 11

2.7 Standards Compliance 11

2.8 Software System Attributes 12

3. Verification 12

4. Supporting Information 12

5. References 12

Revision History

Name	Reason For Changes	Version
Student	Initial draft of CMS Software Requirements Specification	1.0

1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to provide a complete, precise, and verifiable description of the functional and non-functional requirements for the **Complaint Management System (CMS)**. This document serves as a contract between the development team and stakeholders, and acts as a reference throughout the software development lifecycle.

This SRS is structured in accordance with the ISO/IEC/IEEE 29148:2018 standard and is intended for use by software developers, testers, project managers, and academic evaluators.

Note: This is a project submission document for the BTech 3rd Year Software Engineering course.

1.2 Scope

This document specifies requirements for a web-based Complaint Management System that enables registered users to submit, track, and resolve complaints in an organised and transparent manner.

The system shall allow users to:

- Register and log in to the system securely.
- Submit new complaints with category, priority, description, and attachments.
- Track the real-time status of submitted complaints.
- Receive automatic notifications on status changes.
- View complaint history and resolution details.

The system shall allow staff/agents to:

- View assigned complaints and update their status.
- Add internal notes and communicate with the complainant.
- Escalate complaints to higher authorities when required.
- Close complaints after resolution is confirmed.

The system shall allow administrators to:

- Manage user accounts and roles.
- Define complaint categories and priority levels.
- Assign complaints to appropriate departments or agents.
- Generate reports and dashboards on complaint metrics.
- Configure escalation rules and SLA (Service Level Agreement) timelines.

Note: The system runs as a web application accessible through modern browsers. It does not replace existing legal or official complaint channels.

1.3 Product Perspective

1.3.1 System Interfaces

The CMS operates as a standalone web application hosted on a server. It interfaces with a relational database (MySQL / PostgreSQL) for persistent storage, and an SMTP mail server for sending email notifications. The application runs in the latest versions of Chrome, Firefox, Edge, and Safari browsers.

1.3.2 User Interfaces

The application provides a responsive web-based GUI with dashboards, forms, tables, status indicators, and navigation menus. The interface is designed to be intuitive for both technical and non-technical users. Mobile-responsive design ensures usability on smartphones and tablets.

1.3.3 Hardware Interfaces

The system does not require any dedicated hardware beyond a standard web server and client devices (PC, laptop, smartphone) capable of running a modern web browser.

1.3.4 Software Interfaces

The application interfaces with the following software components:

- Relational Database (MySQL 8.0+ or PostgreSQL 14+) for data storage.
- SMTP server (e.g., SendGrid, Gmail SMTP) for email notifications.
- REST API endpoints for integration with third-party reporting tools.
- Optional: SMS gateway API (e.g., Twilio) for SMS notifications.

1.3.5 Communications Interfaces

The application communicates over HTTPS (TLS 1.2 or higher). All API communication uses JSON format. Email notifications follow the SMTP protocol.

1.3.6 Memory Constraints

The server hosting the application shall have a minimum of 4 GB RAM. The database shall support storage of at least 100,000 complaint records with associated attachments up to 10 MB each.

1.3.7 Operations

The system shall be available 24x7 with a minimum uptime of 99.5%. Scheduled maintenance windows shall be communicated to users in advance. Regular automated database backups shall be performed daily.

1.3.8 Site Adaptation Requirements

The system shall support configuration of organisation name, logo, complaint categories, departments, and SLA rules to adapt to different deployment environments without code changes.

1.4 Product Functions

The primary functions of the Complaint Management System are summarised below. Detailed functional requirements are specified in Section 2.2.

- User registration, authentication, and role-based access control.
- Complaint submission with category, description, priority, and file attachments.
- Automated complaint ID generation and acknowledgement.
- Complaint assignment to departments or individual agents.
- Status tracking and lifecycle management (Open → In Progress → Resolved → Closed).
- Automatic escalation based on SLA breach rules.
- Email and/or SMS notifications at key lifecycle events.
- Feedback and rating submission by complainants after resolution.
- Admin dashboard with complaint metrics, charts, and export capabilities.
- Audit trail and history log for every complaint.

1.5 User Characteristics

1.6 Limitations

- The system does not provide real-time chat or live video support.
- The system does not integrate with legacy complaint systems unless custom adapters are developed.
- Offline use is not supported; an active internet connection is required.
- File attachment size is limited to 10 MB per complaint.
- The system does not guarantee legal compliance with complaint resolution laws unless configured appropriately by the administrator.

1.7 Assumptions and Dependencies

- It is assumed that all users have a stable internet connection and a modern web browser.
- The organisation deploying the system will configure complaint categories, departments, and SLA timelines before go-live.
- The system depends on an operational SMTP server for email notifications.
- User authentication depends on a secure password storage mechanism (bcrypt hashing).
- The deployment environment provides HTTPS certificates for secure communication.

1.8 Definitions

Complaint: A formal expression of dissatisfaction submitted by a user regarding a product, service, or experience.

SLA (Service Level Agreement): A defined time limit within which a complaint must be acknowledged, addressed, or resolved.

Escalation: The process of forwarding an unresolved complaint to a higher authority when SLA timelines are breached.

Agent: A staff member responsible for handling and resolving complaints assigned to them.

Status: The current stage of a complaint in its lifecycle (e.g., Open, In Progress, Resolved, Closed).

Category: A predefined classification grouping complaints of similar nature (e.g., Billing, Technical, Service).

Priority: A level indicating the urgency of a complaint (e.g., Low, Medium, High, Critical).

Audit Trail: A chronological record of all actions taken on a complaint, including user, timestamp, and action description.

1.9 Acronyms and Abbreviations

Acronym	Full Form

API	Application Programming Interface
CMS	Complaint Management System
CSV	Comma Separated Values
GUI	Graphical User Interface
HTML	Hypertext Markup Language
HTTP/HTTPS	Hypertext Transfer Protocol / Secure
JSON	JavaScript Object Notation
REST	Representational State Transfer
SLA	Service Level Agreement
SMTP	Simple Mail Transfer Protocol
SRS	Software Requirements Specification
UI	User Interface
UX	User Experience

2. Requirements

2.1 External Interfaces

The CMS exposes a RESTful API for integration with third-party tools. External interfaces include the email notification service (SMTP), optional SMS gateway, and the browser-based frontend communicating with the backend over HTTPS.

[CMS-SRS-01]	The application shall provide a RESTful API with JSON payloads for all data exchange between the frontend and backend.
[CMS-SRS-02]	The application shall communicate over HTTPS using TLS 1.2 or higher for all client-server interactions.
[CMS-SRS-03]	The application shall interface with an SMTP server to send email notifications to users and agents.

2.2 Functions

2.2.1 User Registration and Authentication

2.2.1.1 User Registration

[CMS-SRS-10]	The application shall allow new users to register by providing their full name, email address, mobile number, and a password.
[CMS-SRS-11]	The application shall validate the email address during registration by sending a verification link to the registered email.
[CMS-SRS-12]	The application shall enforce a minimum password length of 8 characters including at least one uppercase letter, one digit, and one special character.
[CMS-SRS-13]	The application shall prevent registration with an email address that is already associated with an existing account.

2.2.1.2 User Login

[CMS-SRS-14]	The application shall allow registered users to log in using their email address and password.
[CMS-SRS-15]	The application shall lock a user account for 15 minutes after 5 consecutive failed login attempts and notify the user via email.
[CMS-SRS-16]	The application shall support a 'Forgot Password' feature allowing users to reset their password via a secure time-limited link sent to their registered email.
[CMS-SRS-17]	The application shall maintain authenticated sessions using secure HTTP-only cookies or JWT tokens and invalidate sessions after 30 minutes of inactivity.

2.2.1.3 Role-Based Access Control

[CMS-SRS-18]	The application shall support three user roles: Complainant, Agent, and Administrator.
[CMS-SRS-19]	The application shall restrict access to features based on the user's assigned role.
[CMS-SRS-20]	The application shall allow Administrators to assign or revoke roles for any user account.

2.2.2 Complaint Submission

2.2.2.1 Create Complaint

[CMS-SRS-21]	The application shall allow authenticated Complainants to submit a new complaint by filling a form with the following mandatory fields: title, description, category, and priority level.
[CMS-SRS-22]	The application shall automatically assign a unique alphanumeric Complaint ID to each submitted complaint (e.g., CMS-2026-00142).
[CMS-SRS-23]	The application shall record the submission date and time, user ID, and IP address for each complaint.
[CMS-SRS-24]	The application shall allow complainants to attach up to 5 files (images, PDFs, documents) with a maximum size of 10 MB per file to a complaint.
[CMS-SRS-25]	Upon successful submission, the application shall send an automated acknowledgement email containing the Complaint ID and expected resolution timeline to the complainant.
[CMS-SRS-26]	The application shall set the initial status of all newly submitted complaints to 'Open'.

2.2.2.2 Complaint Categories and Priority

[CMS-SRS-27]	The application shall allow Administrators to create, edit, and delete complaint categories (e.g., Technical, Billing, Service Quality, Infrastructure).
[CMS-SRS-28]	The application shall support four priority levels: Low, Medium, High, and Critical.
[CMS-SRS-29]	The application shall allow Administrators to define SLA response times for each combination of category and priority level.

2.2.3 Complaint Tracking

2.2.3.1 View Complaint Status

[CMS-SRS-30]	The application shall allow Complainants to view the current status and full history of their submitted complaints through a personal dashboard.

[CMS-SRS-31]	The application shall display the complaint lifecycle as a visual status timeline showing the stages: Open → In Progress → Resolved → Closed.
[CMS-SRS-32]	The application shall allow Complainants to search for a complaint using its unique Complaint ID.
[CMS-SRS-33]	The application shall allow Complainants to filter their complaints by status, category, priority, and date range.

2.2.3.2 Audit Trail

[CMS-SRS-34]	The application shall maintain a complete audit trail for each complaint recording every status change, comment addition, assignment, and escalation with the responsible user's name and timestamp.
[CMS-SRS-35]	The application shall display the audit trail to the assigned Agent and Administrator. Complainants shall only see public-facing updates.

2.2.4 Complaint Assignment and Escalation

2.2.4.1 Assignment

[CMS-SRS-36]	The application shall allow Administrators to manually assign any Open complaint to a specific Agent or department.
[CMS-SRS-37]	The application shall support automatic assignment of complaints to Agents based on predefined round-robin or category-based routing rules.
[CMS-SRS-38]	The application shall notify the assigned Agent via email when a new complaint is assigned to them.
[CMS-SRS-39]	The application shall allow Agents to reassign a complaint to another Agent with a mandatory reason note.

2.2.4.2 Escalation

[CMS-SRS-40]	The application shall automatically escalate a complaint to the Administrator if the defined SLA resolution time is breached without the complaint being resolved.
[CMS-SRS-41]	Upon escalation, the application shall notify both the Administrator and the Complainant via email with the escalation reason.
[CMS-SRS-42]	The application shall allow Agents to manually escalate a complaint with a mandatory escalation reason.
[CMS-SRS-43]	The application shall display an escalation badge on all escalated complaints in the Administrator dashboard.

2.2.5 Resolution and Closure

[CMS-SRS-44]	The application shall allow Agents to mark a complaint as 'Resolved' by providing a resolution description and optionally attaching supporting documents.
[CMS-SRS-45]	Upon marking a complaint as Resolved, the application shall notify the Complainant via email with the resolution details and a link to submit feedback.
[CMS-SRS-46]	The application shall allow Complainants to either accept the resolution (which sets status to 'Closed') or reject it with a reason (which re-opens the complaint).
[CMS-SRS-47]	The application shall automatically close a Resolved complaint if the Complainant does not take any action within 7 calendar days.
[CMS-SRS-48]	The application shall allow Complainants to submit a satisfaction rating (1–5 stars) and optional feedback comment upon complaint closure.

2.2.6 Notifications

[CMS-SRS-49]	The application shall send email notifications to the Complainant at each of the following events: complaint submitted, complaint assigned, status updated, complaint escalated, and complaint resolved/closed.
[CMS-SRS-50]	The application shall send email notifications to the Agent at the following events: new complaint assigned, complaint re-opened, and complaint escalated.
[CMS-SRS-51]	The application shall display in-app notification badges for unread updates within the user's dashboard.
[CMS-SRS-52]	The application shall allow users to configure their notification preferences to enable or disable email notifications for specific event types.

2.2.7 Admin and Reporting

2.2.7.1 Admin Dashboard

[CMS-SRS-53]	The application shall provide Administrators with a dashboard displaying key metrics: total complaints, open complaints, resolved complaints, average resolution time, and SLA breach count.
[CMS-SRS-54]	The Administrator dashboard shall display a bar chart of complaints by category and a line chart of complaint volume over time.
[CMS-SRS-55]	The application shall allow Administrators to view all complaints in a filterable and sortable table with columns: Complaint ID, Title, Category, Priority, Status, Assigned Agent, and Submission Date.

2.2.7.2 Report Generation

[CMS-SRS-56]	The application shall allow Administrators to generate complaint reports filtered by date range, category, status, agent, and priority.

[CMS-SRS-57]	The application shall allow Administrators to export complaint reports to PDF and CSV formats.
[CMS-SRS-58]	The application shall generate an SLA compliance report showing the percentage of complaints resolved within the defined SLA for each category.

2.2.7.3 User Management

[CMS-SRS-59]	The application shall allow Administrators to create, view, edit, deactivate, and delete user accounts.
[CMS-SRS-60]	The application shall allow Administrators to assign or revoke roles for any user.
[CMS-SRS-61]	The application shall log all Administrator actions (user creation, deletion, role changes) in a system audit log.

2.3 Usability Requirements

[CMS-SRS-62]	The application shall have an intuitive interface such that a new Complainant can successfully submit their first complaint within 5 minutes without any training.
[CMS-SRS-63]	The application shall be fully responsive and usable on screen widths ranging from 320px (mobile) to 2560px (large desktop).
[CMS-SRS-64]	The application shall provide clear error messages with corrective guidance whenever a form submission fails due to validation errors.
[CMS-SRS-65]	The application shall follow WCAG 2.1 Level AA accessibility guidelines to support users with disabilities.
[CMS-SRS-66]	The application shall support English as the primary language, with the architecture allowing for future internationalisation (i18n).

2.4 Performance Requirements

2.4.1 Response Time

[CMS-SRS-67]	The application should load the main dashboard within 3 seconds on a standard broadband connection (≥10 Mbps).
[CMS-SRS-68]	The application should complete any form submission or status update operation within 2 seconds under normal load conditions.

2.4.2 Throughput and Scalability

[CMS-SRS-69]	The application shall support at least 500 concurrent users without performance degradation beyond acceptable response time thresholds.
[CMS-SRS-70]	The application shall allow storage and retrieval of at least 1,000,000 complaint records without degradation in query response time beyond 5 seconds.

2.4.3 Availability

[CMS-SRS-71]	The application shall achieve a minimum uptime of 99.5% per month, excluding scheduled maintenance windows.

2.5 Logical Database Requirements

[CMS-SRS-72]	The application shall store all complaint data, user profiles, attachments metadata, audit logs, and configuration settings in a relational database.
[CMS-SRS-73]	The application shall encrypt all sensitive personal data (passwords, mobile numbers, email addresses) at rest using AES-256 encryption.
[CMS-SRS-74]	The application shall perform automated daily backups of the entire database and retain backups for a minimum of 30 days.

[CMS-SRS-75]	The database schema shall enforce referential integrity using foreign key constraints to maintain consistency between complaints, users, categories, and audit records.

2.6 Design Constraints

[CMS-SRS-76]	The backend of the application shall be developed using a widely adopted web framework (e.g., Django, Node.js/Express, Spring Boot).
[CMS-SRS-77]	The frontend shall be implemented using HTML5, CSS3, and JavaScript (or a framework such as React or Vue.js).
[CMS-SRS-78]	The application architecture shall follow the MVC (Model-View-Controller) or equivalent design pattern to ensure separation of concerns.
[CMS-SRS-79]	All source code shall be maintained in a version control system (e.g., Git) with meaningful commit messages and branching strategy.

2.7 Standards Compliance

[CMS-SRS-80]	This SRS document shall comply with the ISO/IEC/IEEE 29148:2018 standard for systems and software engineering requirements specifications.
[CMS-SRS-81]	The application shall comply with applicable data protection regulations (e.g., PDPA, GDPR) regarding collection, storage, and processing of personal data.
[CMS-SRS-82]	The application's REST API shall comply with OpenAPI 3.0 specification for documentation and third-party integration.

2.8 Software System Attributes

2.8.1 Security

[CMS-SRS-83]	The application shall run exclusively over HTTPS and reject any plain HTTP requests by redirecting them to HTTPS.
[CMS-SRS-84]	The application shall implement protection against common web vulnerabilities including SQL Injection, Cross-Site Scripting (XSS), Cross-Site Request Forgery (CSRF), and Broken Access Control (OWASP Top 10).
[CMS-SRS-85]	The application shall sanitize and validate all user inputs on both client and server sides before processing or storing.
[CMS-SRS-86]	The application shall not expose sensitive system information (stack traces, database errors) in user-facing error messages.

2.8.2 Reliability

[CMS-SRS-87]	The application shall handle unexpected errors gracefully and log them to a server-side error monitoring system without crashing.
[CMS-SRS-88]	The application shall preserve all submitted complaint data even in the event of a server restart or unexpected shutdown through transactional database writes.

2.8.3 Maintainability

[CMS-SRS-89]	The application code shall follow consistent coding style guidelines and include inline documentation for all public functions and modules.
[CMS-SRS-90]	The application shall include a comprehensive test suite covering at least 80% of business logic code paths.

2.8.4 Portability

[CMS-SRS-91]	The application shall run on any operating system (Windows, Linux, macOS) that supports the chosen runtime environment.
[CMS-SRS-92]	The application shall be containerisable using Docker to facilitate deployment across different hosting environments.

3. Verification

Verification tests for this system shall be specified in a separate Verification and Testing document [CMS-TESTS]. Each requirement in this SRS is traceable to one or more test cases.

The verification strategy includes the following test types:

- **Unit Tests:** Verifying individual functions and modules in isolation.
- **Integration Tests:** Verifying the interaction between modules (e.g., complaint submission triggering email notification).
- **System Tests:** End-to-end testing of complete user workflows.
- **Performance Tests:** Load and stress testing to validate performance requirements (Section 2.4).
- **Security Tests:** Penetration testing and vulnerability scanning.
- **User Acceptance Tests (UAT):** Validation by end users against the stated requirements.

4. Supporting Information

The following supporting artefacts complement this SRS document and together form the complete project documentation package:

- Data Flow Diagram (DFD) – Level 0 and Level 1 diagrams illustrating system data flows.
- Entity Relationship Diagram (ERD) – Database schema and entity relationships.
- Use Case Diagram – Visual representation of system actors and use cases.
- System Architecture Diagram – High-level deployment and component architecture.
- Wireframes / UI Mockups – Low-fidelity designs for key screens.
- Project Plan / Gantt Chart – Timeline and milestones for the development project.

5. References

Reference	Description
[CMS-NEEDS]	Complaint Management System – User Needs Document
[CMS-TESTS]	Complaint Management System – Verification and Test Cases
[ISO29148]	ISO/IEC/IEEE 29148:2018 – Systems and Software Engineering – Life Cycle Processes – Requirements Eng
[OWASP]	OWASP Top 10 Web Application Security Risks – https://owasp.org/www-project-top-ten/
[WCAG21]	Web Content Accessibility Guidelines (WCAG) 2.1 – https://www.w3.org/TR/WCAG21/
[ReqViewEx]	ReqView SRS Example – Eccam s.r.o. (reference template) – https://www.reqview.com/

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